

Experiment - 04

Name: Samridhi Singh

Section: 24AIT_KRG-1_A

UID: 24BAI70514

Semester: 5th

Branch: BE-CSE(AIML) Subject: Advanced Database Management System

Question 1: Happiness Ranking Reorder

You are given a table `tbl_happiness` that stores the happiness ranking of different countries. (i) Write a query to retrieve all the records from the table while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order. (ii) Modify the query so that all the rankings are displayed in descending order, except for India and Sri Lanka, which must still appear at the top.

tbl_happiness Table:

Column Name	Type
rankings	integer
country	varchar

tbl_happiness Example Input:

rankings	country
1	Finland
2	Denmark
3	Iceland
4	Israel
5	Netherlands
6	Sweden
7	Norway
126	India
128	Sri Lanka

Example Output (i):

rankings	country
126	India
128	Sri Lanka
1	Finland
2	Denmark
3	Iceland
4	Israel
5	Netherlands
6	Sweden
7	Norway

Example Output (ii):

rankings	country
126	India
128	Sri Lanka
7	Norway
6	Sweden
5	Netherlands
4	Israel

3	Iceland
2	Denmark
1	Finland

Solution (i):

```
SELECT
    RANKINGS,
    COUNTRY
FROM TBL_HAPPINESS
ORDER BY
    (COUNTRY IN ('India', 'Sri Lanka')) DESC,
    RANKINGS ASC;
```

Solution (ii):

```
SELECT
    RANKINGS,
    COUNTRY
FROM TBL_HAPPINESS
ORDER BY
    (COUNTRY IN ('India', 'Sri Lanka')) DESC,
    CASE WHEN COUNTRY IN ('India', 'Sri Lanka') THEN RANKINGS END ASC,
    CASE WHEN COUNTRY NOT IN ('India', 'Sri Lanka') THEN RANKINGS END DESC;
```

Question 2: Employees Reporting to a Manager

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number, and the final result should be sorted by employee ID.

Employee Table:

Column Name	Type
employee_id	integer
name	varchar
reports_to	integer
age	integer

Employee Example Input:

employee_id	name	reports_to	age
9	Hercy	NULL	43
6	Alice	9	41
4	Bob	9	36
2	Winston	NULL	37

Example Output:

employee_id	name	reports_count	average_age
9	Hercy	2	39

Solution:

```
SELECT
```

```
M.EMPLOYEE_ID,  
M.NAME,  
COUNT(E.EMPLOYEE_ID) AS REPORTS_COUNT,  
ROUND(AVG(E.AGE)) AS AVERAGE_AGE  
FROM EMPLOYEE AS M  
JOIN EMPLOYEE AS E  
  ON E.REPORTS_TO = M.EMPLOYEE_ID  
GROUP BY M.EMPLOYEE_ID, M.NAME  
ORDER BY M.EMPLOYEE_ID;
```